# Estimation of the number of regular customers using POS data of convenience stores

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Purchase behavior of consumers is recorded in detail in the form of point-of-sales (in short POS) data. We analyze POS data provided by a convenience store chain, in which names of purchased items, prices and the time are recorded for each transaction. We show examples of data analyses to clarify consumers' behavior.

In general, the number of sales of an item in one store is theoretically expected to obey a Poisson distribution, and this assumption is confirmed by the real data for relatively short observation window (typically one hour or less) for popular items which are supplied sufficiently. If we observe the distribution of sales in a longer observation window we find deviation from the Poisson distribution just like the case of blog entry number fluctuations, which is caused by the number changes of potential consumers [1].

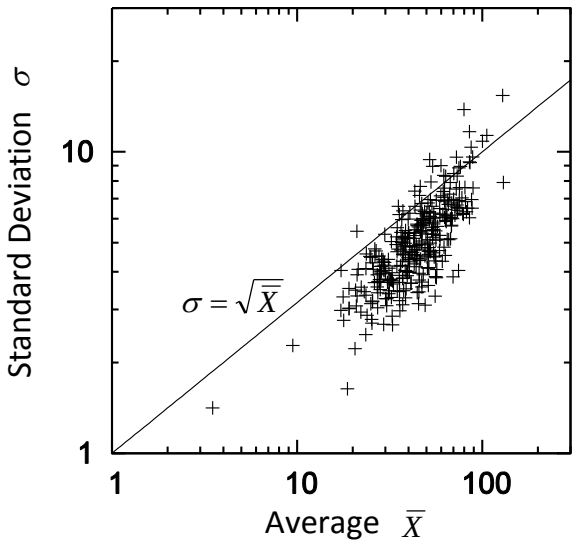



Fig.1: Scatter plot of averages and standard deviations of the number of sales of a magazine at stores.

In this paper we pay attention to the number of sales of specific products such as popular magazines which are sold periodically every week. The standard deviation of sales number fluctuation is smaller than that of simple Poisson distribution by the effect of regular customers as shown in Fig.1. We introduce a method of estimating the number of regular customers from time series of POS data.

［1］Yukie Sano, Kenta Yamada, Hayafumi Watanabe, Hideki Takayasu, Misako Takayasu, "Empirical analysis of collective human behavior for extraordinary events in the blogosphere", **Phys. Rev. E**87, 012805, 20